

Session 4 · Workflows on Live SEC Data

Yesterday you pasted data by hand. Today your code fetches it from the U.S. Securities and Exchange Commission (SEC).

Plan: concepts 15' · live demo 22' · your lab 30' · debrief 8'

By the end of Session 4 you can

1. **Draw** the workflow pattern and defend the code-vs-model split
2. **Pull** real fundamentals from SEC filings with one function call
3. **Build** a screen you can rerun tomorrow, criteria as dials
4. **Audit** the model's own prose: every number traced to an input
5. **Train** a forecasting model and measure its error before trusting it

A workflow is a plan the model cannot change

INPUT → RETRIEVE → STRUCTURE → REASON → VALIDATE → HUMAN
(code) (code) (model) (code) (you)

- **Code does the arithmetic:** pandas is deterministic; the model is not
- **The model reasons once,** in the middle, about a table it is *given*
- **Code audits the prose,** and a human approves: nothing is published without explicit sign-off

EDGAR: every filing, free, no key

- **EDGAR** (Electronic Data Gathering, Analysis and Retrieval) is the U.S. Securities and Exchange Commission's public filing database
- **What it contains:** every 10-K, 10-Q and 8-K since the nineties, machine readable
- **What it does not contain:** market prices and analyst estimates — filings are not market data

Three traps you will meet, because **companies tag their own accounts:**

1. The same item **changes names** across years
2. Foreign filers report in **local currency**
3. Most pharma reports **no operating income line at all**

Demo: a memo with two moments to watch

NVIDIA vs AMD vs Intel, live from the filings.

- **First, the data gaps:** the memo lists what it could *not* know from its inputs — the blind spots are declared, not hidden
- **Second, a planted error:** one figure in the memo is altered by hand, and the audit finds it

audit: every figure in the prose → checked against inputs → the altered figure is flagged

The libraries in this session

Library	Role here	Standing
requests	fetches SEC filings over HTTP	the standard HTTP library
pandas	the deterministic screen	the industry standard for data work
numpy	trains the least-squares forecaster	the numerical foundation of Python
pydantic	typed, validated model output	the industry standard for validation

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
mask = (df[None] >= min_growth) & (df[None] >= min_margin)  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- **Everything else is given.** Do not rewrite it.
- Then run the  **check cell** directly below. Green means correct: continue.

Stuck for two minutes? Select the lines, press **Option+K** (**Alt+K** on Windows), and ask the ***** panel.

Your lab · notebook 04 · 30 minutes

A screening engine: 16 real industrials and pharma, live from EDGAR

- **Exercise 1:** fetch 3 years of fundamentals, tolerating failed tickers
- **Exercise 2:** the filter — **pandas decides which companies pass, never the model**
- **Exercise 3:** one grounded rationale per survivor + numeric audit
- **Exercise 4:** train a revenue forecaster; the measured error is the product

Question to answer before the debrief: why net margin, not operating margin?

Key takeaway

A screen you can rerun tomorrow is worth more than ten analyses performed once.

Next session: the model plans its own steps, under controls you define.